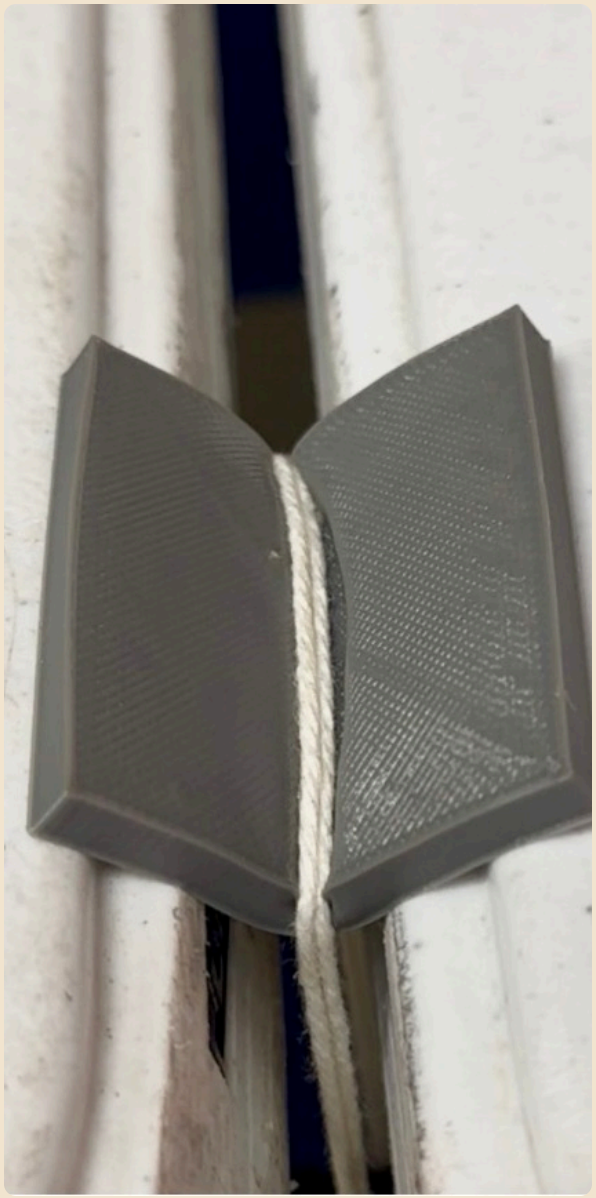
PROJECT OVERVIEW

An astronaut who tears an ankle ligament on the Moon can't wait days for a ride home. OrthoPrint is a splint printed on the lunar base from a scan of the patient's own leg, so it fits, holds the ankle still and goes on with one hand.



Printed prototype plates in the hanging-weight test

OUR PROTOTYPE



Strap, front · Side · Back · Reusable zip tie idea

Printed straps lock like a reusable zip tie and wrap the ankle in the athletic-tape X.

MATERIALS

PLA Prototype 1	**ABS** Prototype 2
TPU Prototype 3	**PETG** Tested

CF-PEEK
Our pick for flight

NEXT STEPS

→ Test strap thickness and break load
→ Find where the filament softens with heat
→ Prototype #2 for the CDR

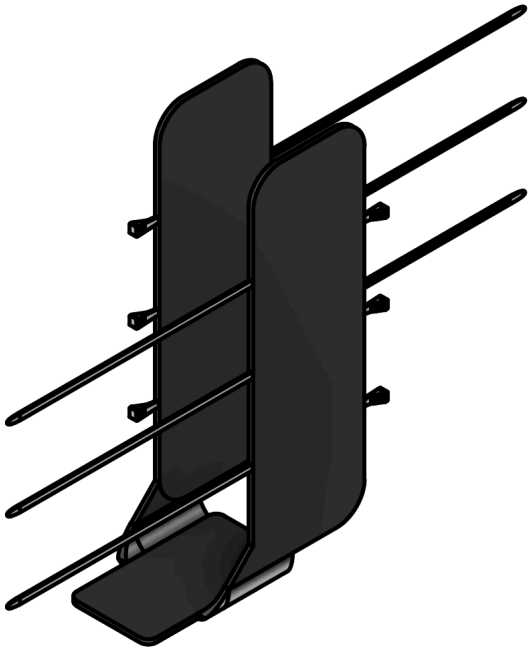
A NASA HUNCH 2026–27 student design project. Not a medical device.

GLADYS PORTER EARLY COLLEGE HIGH SCHOOL
BROWNSVILLE, TX

NASA HUNCH · LUNAR SPLINT DESIGN CHALLENGE

ORTHOPRINT

PDR Design & Prototype · 2026–27



TEAM LUNOVA

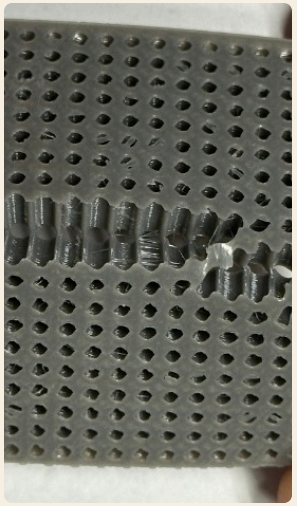
Gertrudis · Jose · Nancy

Teachers: Lupita C. Vasquez and Claudia Cortez

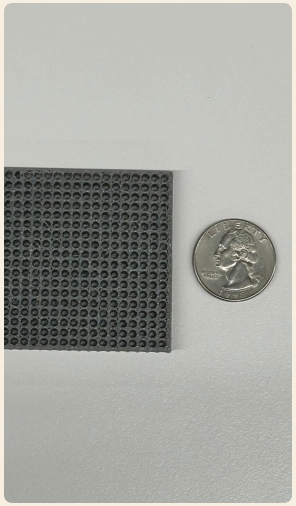
Website, videos & CAD
gertrudish978-pixel.github.io/lunar-splint

HOW IT WORKS

1. **Scan**
   the injured leg
2. **Print**
   the shell and straps on the base
3. **Fit**
   it on vertically
4. **Strap**
   it in an X around the ankle



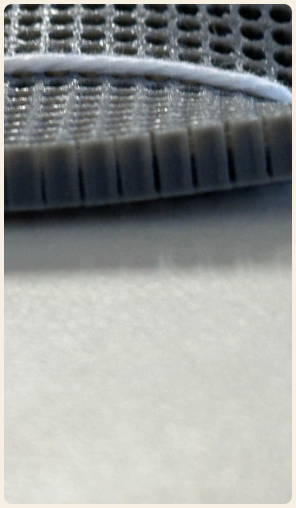
Holes split, so the wall went solid



TPU sample



Load test



Solid TPU, our choice

BUILT FOR THE MOON

SINGLE-PIECE RATCHET CLOSURE

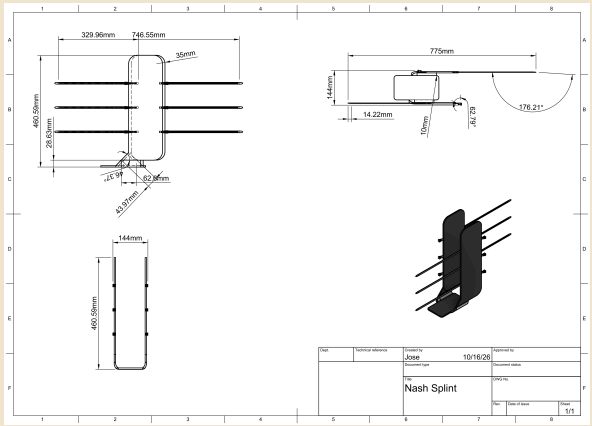
Tightens or releases with one hand. No loose parts.

INTEGRATED HYGIENE

Comes off to clean and check the skin.

THERMAL PHASE DESIGN

Flexible to put on, firm to hold.



Detail drawing (mm) · Jose

WHAT WE BRING

→ A custom fit for every patient, not fixed sizes

→ Straps the astronaut can tighten alone

→ A flight material chosen: CF-PEEK

BY THE NUMBERS

3 lunar innovations, where the challenge asks for one

4 materials tested: PLA, ABS, TPU and PETG

56 students answered our Needs Assessment Survey

63% said the ankle is the most injured limb

54% struggle to find a splint or brace that fits